

Model	Method	Cancer-Myth		CREPE	
		TPR $\uparrow$	FPR $\downarrow$	TPR $\uparrow$	FPR $\downarrow$
Qwen2.5 7B	Direct gate	7.0 ^H	10.7 ^H	—	—
	Review CoT gate (2-step)	29.5 ^H	16.1 ^H	—	—
	Direct + scope protection	—	—	—	—
	Probe gate (hidden states)	75.0 ^H	47.7 ^H	—	—
	GEPA	—	—	—	—
	Ours	—	—	—	—
GPT-6 Luna	Direct gate	—	—	—	—
	Review CoT gate (2-step)	—	—	—	—
	Direct + scope protection	—	—	—	—
	GEPA	—	—	—	—
	Ours	—	—	—	—
Gemma 4 12B	Direct gate	40.5 ^H	16.1 ^H	—	—
	Review CoT gate (2-step)	51.3 ^H	32.9 ^H	—	—
	Direct + scope protection	—	—	—	—
	Probe gate (hidden states)	—	—	—	—
	GEPA	—	—	—	—
	Ours	—	—	—	—
Qwen3.8 27B OFF	Direct gate	26.1 ^H	16.1 ^H	—	—
	Review CoT gate (2-step)	57.5 ^H	35.6 ^H	—	—
	Direct + scope protection	—	—	—	—
	Probe gate (hidden states)	—	—	—	—
	GEPA	—	—	—	—
	Ours	—	—	—	—
Qwen3.8 27B ON	Direct gate	44.1 ^H	21.5 ^H	—	—
	Review CoT gate (2-step)	61.4 ^H	38.3 ^H	—	—
	Direct + scope protection	—	—	—	—
	Probe gate (hidden states)	—	—	—	—
	GEPA	—	—	—	—
	Ours	—	—	—	—
Shared	TF-IDF text	86.1 ^H	51.0 ^H	—	—

Table 1. False-premise detection across models and datasets

Stored results; — means unavailable or not comparable. Values are percentages, one run.
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.
TPR $\uparrow$: FPQ detection; FPR $\downarrow$: false positives on normal questions.
Probes require hidden states; no native Luna probe. TF-IDF is model-independent.
Default crossfit probe/TF-IDF decisions; these are not matched-FPR thresholds.
Ours denotes the proposed method; its implementation is not fixed.